

PBTA Analysis – Summary Report

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1 Introduction

2 Data Processing and Methods

3 Results

in tf vs os_status we see high tf has (lower) death rate which is wierd. the km plot show differently. this might be due to the fact alot of samples with high tf (1) has lower death rate (why?) but at high tf ratios (0.7-0.9) there is high death, and the tf=1 is baising the analysis. we should separate tf =1 samples and see the results.

Questions like what cancer group is interesting or what is interesting in the data are chal-laneging questions and could be tackled in few ways. A simple yet powerfull approach would be to tackle groups that has strong statistics to work on (number of samples and so on). Picking these groups and conducting an analysis on them would be reproducible among other groups and can reveal novel outcomes.

Never the less, to answer these questions I conducted a systematic "unsupervised" approach where I assumed no prior knowledge, or bais on the data. I wanted to highlight interesting variables (clinical, genomic) that can shed light on groups/sub groups in the data. To do so i first investigated the clinical metadata by visulatizing and statisticaly quesionting is variable X (Age, Sex, Tf...) differs in Cancer groups and is it related to death/event status. I evaluated X combinations and found that: cancer groups yyy, differs in variables „,. I further conducted unsupervised clustering techniques on these variables to explore the data structure and found: Second, I utilized the mRNA expression matrix and tried to identify correlated, mutual exclusive, and co-occurring genes. I further tried to connect this insights into my prevoius findings on the clinical data and reveal novel sub groups among patients. I found that the genes: X,Y,Z are in.. A further inspection about the rule of these genes, thier biological pathways is needed.

4 Conclusions and Future Directions

5 References